

Worksheet 5

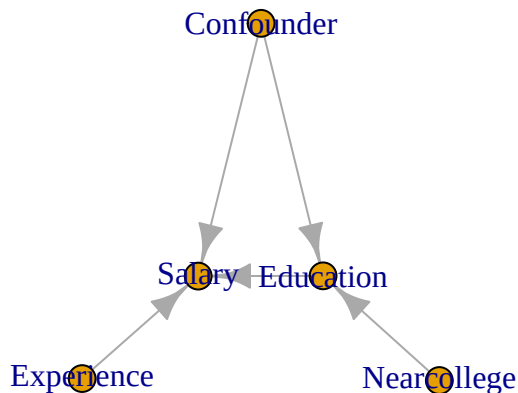
Exercise 1: Neighborhood of schools

The data frame `SchoolingReturns` in the file `schoolingreturns.rda` contains a part of the observational data from a U.S. national longitudinal survey of young men conducted in 1976. The aim of this exercise is to estimate the causal effect of education (`education` = education in 1976 in years) on salary (`wage` = raw wages in 1976 in cents per hour). We assume that there is an unobserved confounding factor, such as motivation, that has a causal effect on education and salary. In addition, there are two more variables: `experience` (years of labor market experience) and `nearcollege` (did the individual grow up near a college).

```
load("schoolingreturns.rda")
```

- a) Draw a DAG that visualizes the assumed causal relationships between the 4 variables given in the data.frame `SchoolingReturns` and the assumed confounder. Give reasons why the neighborhood to a college might be an appropriate instrument for education.

```
set.seed(324)
library(gRbase)
d <- dag(c("Education", "Nearcollege", "Confounder"),
        c("Salary", "Education", "Confounder"),
        c("Salary", "Experience"))
plot(d)
```



The closer one lives to a colleges the less effort it takes to visit a college and hence the more likely is it to visit a college and have a long education.

- b) Check if the variable `nearcollege` can be used as instrument for education. Use an appropriate visualization and test, to confirm that the variable `nearcollege` and `education` are associated.

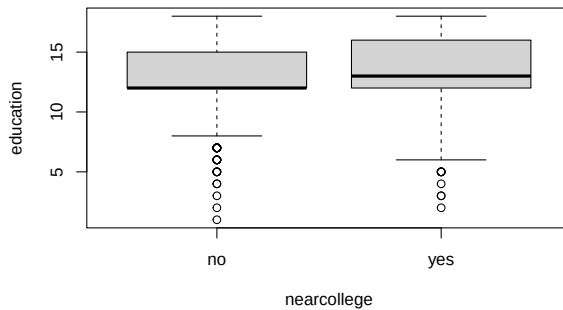
The variable `nearcollege` needs to satisfy the following properties:

1. `nearcollege` need to affect education.
2. `nearcollege` is independent of the confounder (e.g. motivation).

3. `nearcollege` needs to affect salary only through education.

Here, we focus on (1), since the other two we can only argue in favor of them.

```
boxplot(education ~ nearcollege, data = SchoolingReturns)
```



```
t.test(education ~ nearcollege, data = SchoolingReturns)
```

Welch Two Sample t-test

data: education by nearcollege

t = -7.7692, df = 1740.6, p-value = 1.34e-14

alternative hypothesis: true difference in means between group no and group yes is not equal to 0

95 percent confidence interval:

-1.0383043 -0.6197336

sample estimates:

mean in group no	mean in group yes
12.69801	13.52703

The visualization and the t-test show that education and nearcollege are dependent.

- c) Fit a naive regression model taking the log of `wage` as outcome and `education` as explanatory variable. Determine the estimated effect of education on log-wage and explain why this effect might not correspond to the causal effect of education on log-wage.

Transformation:

```
SchoolingReturns$lwage <- log(SchoolingReturns$wage)
```

Regression:

```
fit_naive_unadj <- lm(lwage ~ education, data = SchoolingReturns)
```

```
coef(fit_naive_unadj)["education"]
```

```
education
0.05209424
```

Since its an observational study there might exist unobserved confounder that affect education and wage (i.e. the confounder motivation). In this case, an unadjusted regression model yields a biased causal effect.

- d) Use the function `ivreg` to fit an IV-regression model adding `nearcollege` as instrument for education. Compare and discuss the estimated effect of education on wage.

IV-Regression:

```
library(ivreg)
```

```
fit_unadj <- ivreg(lwage ~ education | nearcollege, data = SchoolingReturns)
```

```
coef(fit_unadj)["education"]
```

```
education
```

0.1880626

The IV-regression estimates the causal effect of education on lwage about 3 times larger than the naive regression of c). The effect of the IV-regression is probably closer to the true causal effect, if nearcollege is a valid instrument.

e) Reproduce the result from d) with a 2-step-least-square (SLS).

```
# Step 1: Estimate the effect of nearcollege on education
step1 <- lm(education ~ nearcollege, data = SchoolingReturns)
hat_education <- fitted(step1)
# Step 2: Use the only predicted values of effort
step2 <- lm(lwage ~ hat_education, data = SchoolingReturns)
coef(step2)
```

```
(Intercept) hat_education
3.7674716    0.1880626
```

Step 2 returns the same estimated causal effect.

e) Now add experience as additional explanatory variable.

Note: Since experience has no instrument, you need to use experience itself as instrument in the ivreg function.

```
# IV-Regression with adjustment experience:
fit_adj <- ivreg(lwage ~ education + experience |
  nearcollege + experience,
  data = SchoolingReturns)
coef(fit_adj)
```

```
(Intercept)  education  experience
1.7949817    0.2620435    0.1119278
```

f) Repeat the IV-regression from e) by doing 2SLS manually.

In step 1 each predictor is modeled by its instrument. If a predictor has no instrument the predictor is used as instrument and the prediction of this predictor is equal to the predictor.

```
# Step 1:
step1_edu <- lm(education ~ nearcollege + experience,
  data = SchoolingReturns)
hat_education <- fitted(step1_edu)
hat_experience <- SchoolingReturns$experience
# Step 2:
step2 <- lm(lwage ~ hat_education + hat_experience,
  data = SchoolingReturns)
coef(step2)
```

```
(Intercept) hat_education hat_experience
1.7949817    0.2620435    0.1119278
```

Exercise 2: Cigarette Demand in the United States

The CigaretteDemand dataset in the R-Package ivreg contains information on cigarette demand in the 48 contiguous U.S. states for the years 1985 and 1995. It includes variables such as per

capita consumption, price, income, and taxes to analyze the economic determinants of cigarette consumption.

- a) Fit a linear regression with the response $\log(\text{packs})$ and the two explanatory variables $\log(\text{rprice})$ and $\log(\text{rincome})$.

```
library(ivreg)
data("CigaretteDemand")
cig_ols <- lm(log(packs) ~ log(rprice) + log(rincome),
              data = CigaretteDemand)
coef(cig_ols)
```

```
(Intercept)  log(rprice) log(rincome)
10.3420288   -1.4065004    0.3438501
```

The regression suggests that a 1% increase in prices is associated with a reduction in cigarette consumption of approximately 1.4%.

- b) Fit an IV regression model (using logarithmic transformations as in part a), with **salestax** used as an instrument for the price. Discuss the estimated effect.

```
cig_iv <- ivreg(log(packs) ~ log(rprice) + log(rincome) |
                salestax + log(rincome),
                data = CigaretteDemand)
cbind(OLS = coef(cig_ols), IV = coef(cig_iv))
```

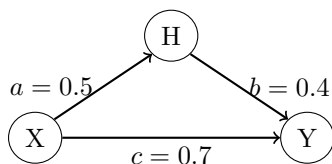
```
              OLS              IV
(Intercept) 10.3420288  9.4306583
log(rprice) -1.4065004 -1.1433751
log(rincome) 0.3438501  0.2145153
```

A 1% increase in prices reduces cigarette consumption by about 1.1%. The estimate may not be reliable despite the use of the IV method: there could be bias due to other unobserved variables.

Exercise 3: Exam Score

A school has set up a supervised remedy program, to give students the opportunity to exercise in addition to their homework. The causal graph for the time spent in the remedy program (X), the time spent with homework (H) and the score achieved in the exam (Y) is depicted below. You can assume a linear SCM (for coefficients see the causal graph) and standard normal distributed noise terms.

(Reference: Judea Pearl et al, *Causal Inference in Statistics - A Primer*)



- a) What is the distribution of the exam score?

The corresponding SCM looks as follows:

$$X \leftarrow E_X$$

$$H \leftarrow 0.5 \cdot X + E_H$$

$$Y \leftarrow 0.4 \cdot H + 0.7 \cdot X + E_Y$$

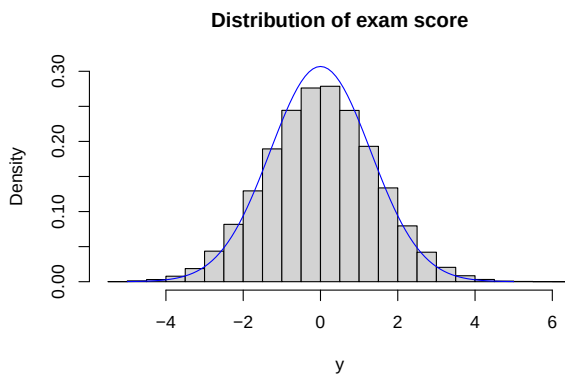
X is normal distributed with $E(X) = 0$ and $Var(X) = 1$.

H is normal distributed with $E(H) = 0.5 \cdot 0 = 0$ and $Var(H) = 0.5^2 \cdot 1 + 1 = 1.25$.

Y is normal distributed with $E(Y) = 0.4 \cdot 0 + 0.7 \cdot 0 + 0 = 0$ and $Var(Y) = 0.4^2 \cdot 1.25 + 0.7^2 \cdot 1 + 1 = 1.69$.

→ The distribution of the exam score is $\mathcal{N}(0, 1.69)$.

```
# This can be checked by simulation
set.seed(456)
N <- 100000
x <- rnorm(N)
h <- 0.5*x + rnorm(N)
y <- 0.4*h + 0.7*x + rnorm(N)
hist(y, freq=FALSE, main = "Distribution of exam score", ylim = c(0,0.3))
lines(seq(-5,5,by=0.1),
      dnorm(seq(-5,5,by=0.1), mean = 0, sd = sqrt(1.69)),
      col="blue")
```



- b) Let us consider a student named Joe, who spent 0.5 hours in the supervised remedy program (i.e. $x = 0.5$) and spent double time for homework (i.e. $h = 1$). In the exam Joe achieved a score of $y = 1.5$. What would Joe's score have been had he doubled his study time of homework?

Abduction:

Determine the noise:

$$e_X = 0.5$$

$$e_H = 1 - 0.5 \cdot 0.5 = 0.75$$

$$e_Y = 1.5 - 0.4 \cdot 1 - 0.7 \cdot 0.5 = 0.75$$

Action:

Modified SCM for $\text{do}(H = 2)$:

$$X \leftarrow E_X$$

$$H \leftarrow 2$$

$$Y \leftarrow 0.4 \cdot H + 0.7 \cdot X + E_Y$$

Prediction:

$$Y_{H=2}^C = 0.4 \cdot 2 + 0.7 \cdot e_X + e_Y = 0.8 + 0.7 \cdot 0.5 + 0.75 = 1.9$$

We conclude that Joe's score had he doubled his homework, would have been 1.9 instead of 1.5.

- c) Let us consider another student named Emily, who spend the same time as Joe in the supervised remedy program (i.e. $x = 0.5$). However, she spent double time as Joe studying (i.e. $h = 2$) and achieved a score of $y = 2.2$ in the exam. What would Emily's score have been had she studied same time as Joe?

Abduction:

Determine the noise:

$$e_X = 0.5$$

$$e_H = 2 - 0.5 \cdot 0.5 = 1.75$$

$$e_Y = 2.2 - 0.4 \cdot 2 - 0.7 \cdot 0.5 = 1.05$$

Action:

Modified SCM same as in b):

Prediction:

$$Y_{H=1}^C = 0.4 \cdot 1 + 0.7 \cdot e_X + e_Y = 0.4 + 0.7 \cdot 0.5 + 1.05 = 1.8$$

We conclude that Emily's score would have been 1.8. She would still have achieved a better result as Joe.

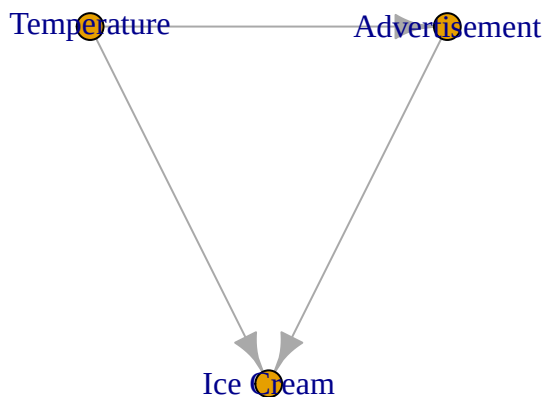
Exercise 4: Ice Cream

Ice cream sales are influenced by various factors. One major factor is temperature. People tend to buy more ice cream in summer and less in winter. Advertisement can also influence how much ice cream is sold. Note that temperature is a driver for advertising ice cream, too.

```
load("ice_cream.rda")
```

- a) Draw a DAG that visualizes the described situation.

```
set.seed(423)
library(gRbase)
d <- dag(c("Advertisement", "Temperature"),
        c("Ice Cream", "Advertisement", "Temperature"))
plot(d)
```



- b) Assume a linear SCM with normal distributed noise and estimate all causal effects from the observed data in the file `ice_cream.rda`.

The three direct causal effects can be estimated using regression models, since a linear SCM is

assumed.

The following adjustments are necessary to estimate the causal effects:

Causal effect of advertising on ice cream: adjustment for temperature is necessary

Causal effect of temperature on ice cream: adjustment for advertising is necessary

Causal effect of temperature on advertising: no adjustment is necessary

Therefore, the following two regressions are needed to estimate the three causal effects:

```
fit1 <- lm(Sales ~ Adv + Temp, data = ice_cream)
coef(fit1)
```

```
(Intercept)      Adv      Temp
-258.251629    2.031745   22.292715
```

```
fit2 <- lm(Adv ~ Temp, data = ice_cream)
coef(fit2)
```

```
(Intercept)      Temp
110.940555      1.740181
```

The estimated SCM is as follows:

$$Temp \leftarrow E_T$$

$$Adv \leftarrow 1.74 \cdot Temp + E_A$$

$$Sales \leftarrow 22.29 \cdot Temp + 2.03 \cdot Adv + E_S$$

c) Have a look at the observation with the highest daily temperature.

- How many ice creams were sold on that day?
- How many ice-creams would have been sold on that particular day, if the temperature would have been only 30?
- How many ice-creams would have been sold on that particular day, if there would have been no advertisement?

```
# Observation with highest temperature
ice_cream[which.max(ice_cream$Temp),]
```

```
Temp      Adv Sales
19 34.1 174.4516   885
```

On the day with the highest temperature 885 ice creams were sold.

Calculating the first counterfactual:

1. Abduction

```
day19 <- ice_cream[19,]
(Es <- day19$Sales - (22.29 * day19$Temp + 2.03 * day19$Adv))
```

```
[1] -229.2258
```

```
(Ea <- day19$Adv - c(1.74 * day19$Temp))
```

```
[1] 115.1176
```

```
(Et <- day19$Temp)
```

```
[1] 34.1
```

2. Action: Modified SCM for $do(Temp = 30)$:

$$Temp \leftarrow 30$$

$$Adv \leftarrow 1.74 \cdot Temp + E_A$$

$$Sales \leftarrow 22.29 \cdot Temp + 2.03 \cdot Adv + E_S$$

3. Prediction

```
temp_c <- 30
adv_c <- 1.74 * 30 + Ea
(sales_c <- 22.29 * temp_c + 2.03 * adv_c + Es)
```

```
[1] 779.129
```

At a temperature of only 30 degrees, 780 ice creams would have been sold.

Calculating the second counterfactual:

1. Abduction is the same as for the first counterfactual.

2. Action: Modified SCM for $\text{do}(Adv = 0)$:

$$Temp \leftarrow E_T$$

$$Adv \leftarrow 0$$

$$Sales \leftarrow 22.29 \cdot Temp + 2.03 \cdot Adv + E_S$$

3. Prediction

```
temp_c <- Et
adv_c <- 0
(sales_c <- 22.29 * temp_c + 2.03 * adv_c + Es)
```

```
[1] 530.8632
```

Without advertising, 531 ice creams would have been sold.

d) Use linear regression to calculate a usual prediction for the two data point described as counterfactual in c). Explain the difference.

```
x0 <- data.frame(Temp = c(30, day19$Temp),
                  Adv = c(day19$Adv, 0))
predict(fit1, newdata = x0)
```

```
      1      2
764.971 501.930
```

The prediction describes the average number of ice creams sold for days with the given temperature and the given amount of advertisement. In contrast, the counterfactual describes how many ice creams would have been sold on the specific day (corresponding to data point 19) if the temperature and advertising had been different.